

WEEK-1

OUT-PUT EXERCISE-2

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:E:\IntelliJ2026\IntelliJ IDEA 2026.1.2\lib\idea_rt.jar=4976
Linear Search:
-----Product found-----
104 | Watch | Accessories

Binary Search:
-----Product found-----
104 | Watch | Accessories
Linear Search:
Product not found

Binary Search:
Product not found

Process finished with exit code 0
```

Explanation:

Time Complexity Comparison

Algorithm	Best Case	Average Case	Worst Case
Linear Search	$O(1)$	$O(n)$	$O(n)$
Binary Search	$O(1)$	$O(\log n)$	$O(\log n)$

Suitable Algorithm for an E-commerce Platform

Binary Search is more suitable for an e-commerce platform because:

- E-commerce websites usually store a very large number of products.
- Binary search significantly reduces search time.
- It works efficiently on sorted product data.
- Faster searches improve user experience and reduce server load.

However, if products are frequently added or removed and maintaining a sorted list is difficult, linear search may be used for small datasets. For large-scale platforms, binary search (or more advanced indexing techniques) is generally preferred.